Susanta Khamrui

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Medinipur, West Bengal, India

Professional Profile

M.Tech in Data Science (PDPM IIITDM Jabalpur, graduated June 2026), with a B.Tech in Electronics and Communication Engineering. My research centers on attention-driven, multi-modal fusion architectures for medical image analysis, with emphasis on segmenting and localizing small, visually subtle structures such as early-stage colorectal polyps. Proficient in Python, C++, and PyTorch, with hands-on experience adapting large-scale foundation models via frozen-encoder and parameter-efficient strategies to learn reliably from limited, noisy data – a constraint common across specialized computer vision domains, not only clinical imaging. Motivated to extend this foundation toward explainable, robust computer vision systems trustworthy enough for deployment in high-stakes settings.

Technical Interests

- **Medical & Clinical Computer Vision:** Segmentation, localization, and detection architectures for clinical imaging data, with emphasis on small, visually subtle targets.
- **Attention-Driven & Multi-Modal Fusion Architectures:** Designing neural networks that combine heterogeneous feature streams into unified, task-relevant representations.
- **Explainable & Trustworthy AI:** Building interpretability into model architectures directly – via attention mechanisms whose weights are themselves evidence – rather than adding explanation methods after training.
- **Foundation Model Adaptation & Resource-Efficient Deep Learning:** Adapting large pretrained visual encoders under tight compute and data constraints typical of specialized, data-scarce domains.

Education

- **PDPM IIITDM Jabalpur** — **August 2024 – June 2026**
  M.Tech *in Computer Science and Engineering (Data Science Specialization)* — *CGPA: 8.7 / 10*
  Thesis: Representation Learning via Hybrid Fusion Architectures for Visual Segmentation

- **Dr. B. C. Roy Engineering College, Durgapur (MAKAUT)** — **August 2019 – June 2023**
  B.Tech *in Electronics and Communication Engineering* — *CGPA: 8.5 / 10*
  Focus: Digital Signal Processing, Probability & Statistics, and Engineering Mathematics

Research & Project Experience

- **Attention-Driven Fusion Architectures for Medical Image Segmentation (M.Tech Thesis)**
  Supervisor: Dr. Ayan Seal, PDPM IIITDM Jabalpur — *February 2025 – June 2026*
  - Engineered “TAP-FuseNet,” a hybrid architecture using Transformer attention to fuse multiple feature streams for segmenting small, visually subtle targets (e.g. early-stage polyps) that standard models under-detect, and adapted large foundation models via frozen-encoder strategies to keep compute costs low.
  - Result: hands-on expertise in attention-driven, multi-modal fusion architectures that generalize under noisy, real-world conditions – transferable across clinical imaging and other data-constrained detection problems.

- **Visual Saliency & Attention Mechanisms**
  Research Assistant, PDPM IIITDM Jabalpur — *August 2024 – January 2025*
  - Implemented multi-scale attention mechanisms and pyramidal hierarchies so models focus computation on relevant regions rather than background, cutting unnecessary processing overhead.
  - Result: learned to build models whose internal attention is itself a source of explanation, and to reason about model behaviour under noisy or corrupted inputs – core to both explainability and robustness in applied computer vision.

Publications

- S. Khamrui and A. Seal, *ADPNet: An Attention-Driven Polyp Segmentation Network with Statistical Feature Extraction.*
  7th International Conference on Frontiers in Computing and Systems (COMSYS), 2026. (Accepted)

- S. Khamrui, A. Seal, A. K. Gupta, M. Penhaker, and O. Krejcar, *An Instrument for Small Polyp Localization in Colonoscopy Images using a Task-Aware Progressive Fusion Network.*
  IEEE Transactions on Circuits and Systems for Video Technology (Under Review)

- S. Khamrui, A. Seal, M. Penhaker, and O. Krejcar, *BAHNet: A Boundary-Aware Hybrid Network for Small Polyp Segmentation in Colonoscopy*, 2026.
  IEEE Journal of Biomedical and Health Informatics (Under Review)

Technical Skills

- **Languages & Tools:** C++, Python, Java, Git/GitHub, Linux, LaTeX, VS Code.
- **Machine Learning & DL:** PyTorch, Attention Mechanisms, Multi-Modal Fusion, Foundation Model Adaptation, Transformers, Medical Image Segmentation.
- **Signal & Image Processing:** DSP, NumPy, SciPy, Pandas.

Language Proficiency

English — Fluent (Professional)   Bengali — Native   Hindi — Fluent

Referees

Dr. Ayan Seal (Supervisor)
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Dr. Sraban Kumar Mohanty
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